SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO3_AT1 — EXPERIMENT

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO3 – Construction (BL3)	**Assessment:**	Assessment Tool 1 – Experiment
Weightage:	20% of CIA	**Total Marks:**	25 (5 Questions × 5 Marks)
CO3 Statement:	*Apply appropriate experimental charting techniques — pie charts, bean plots, violin plots, box plots, and stacked bar charts — to construct accurate visualizations from given data.(BL3)*		
Student Name:	RAGUL.M	**Reg. No.:**	192524066
Section / Batch:	SLOT A	**Date:**	11/08/2026

Code of Conduct

I RAGUL.M 192524066 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: RAGUL.M

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 75 minutes.
- Graph paper or a ruler is recommended for accurate, to-scale drawing.
- Show all calculations (e.g., pie chart angles, five-number summary, IQR) as part of your answer.
- Every chart must include a title, correctly labeled axes (where applicable), and a legend where relevant.

Annexure A – EXPERIMENT

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT1 — Experiment

CO3 · BT3: Apply ·

This rubric is used to assess student responses to the CO3-AT1 Experiment, in which students are given a small dataset and must construct an accurate pie chart, bean plot, violin plot, box plot, or stacked bar chart.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough, accurate understanding of the correct chart type (pie, bean, violin, box, or stacked bar) and the elements required to construct it for the given data.	Good understanding of the required chart type and construction elements, with only minor gaps.	Basic understanding of chart construction, but with some misconceptions about the correct chart type or required elements.	Poor understanding of core construction concepts; selects a chart type fundamentally unsuited to the data.
Explanation	**25%**	Shows detailed, clearly worked calculations (e.g., pie chart angles, five-number summary, IQR, stacked totals) and explains each construction step.	Shows clear calculations and construction steps, with adequate explanation for most parts.	Calculations or construction steps are shown but incomplete, or explanations are superficial.	Little or no working shown; calculations and construction steps are missing or unexplained.
Application	**20%**	Effectively applies chart construction techniques to produce a complete, accurate chart correctly matched to the data.	Applies construction techniques to produce a correct chart, with only minor errors.	Limited application of construction techniques; the chart partially represents the data but has notable errors.	Fails to apply appropriate construction techniques; the chart does not correctly represent the data.
Organization	**15%**	The chart is neatly and	The chart is well organized, with	Basic construction;	Poorly constructed; the

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		logically constructed, with a title, correctly labeled axes or slices, a legend where applicable, and elements in a clear order.	only minor issues in labeling, legend, or layout.	required elements (title, labels, legend) are present but poorly arranged or partly missing.	chart is missing key elements such as a title, labels, or a legend.
Accuracy	**10%**	The chart is completely accurate — all values, proportions, quartiles, and calculations correctly match the given data.	Mostly accurate, with only minor omissions or a small error in one value or calculation.	Partially correct; some plotted values, proportions, or calculations are missing or incorrect.	Largely inaccurate; major errors in plotted values, proportions, or calculations relative to the given data.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

P4 — PIE CHART: 24-HOUR TIME ALLOCATION

Aim:

To construct a pie chart to represent the distribution of a student's daily 24-hour time among different activities.

Data:

The activities are Sleep – 8 hours, Study – 6 hours, Leisure – 4 hours, and Other – 6 hours. The total time is 24 hours.

Method:

The percentage is calculated using (Activity Hours / 24) × 100, and the angle is calculated using (Activity Hours / 24) × 360°.

Analysis:

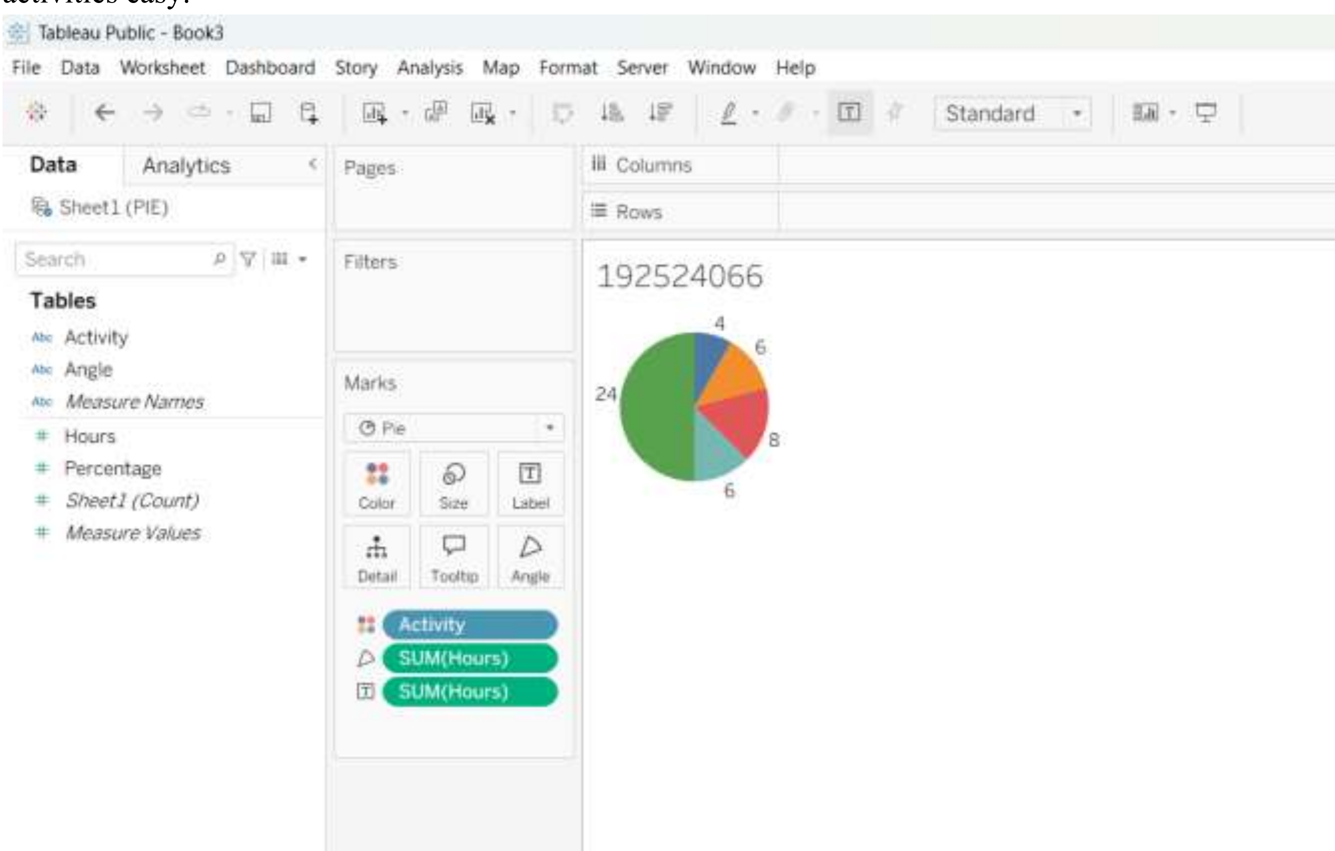
Sleep represents 33.33% with an angle of 120°. Study represents 25% with an angle of 90°. Leisure represents 16.67% with an angle of 60°. Other activities represent 25% with an angle of 90°. The calculated percentages together form 100% and the angles form 360°.

Interpretation:

The pie chart shows that Sleep occupies the largest share of the student's day. Study and Other activities have equal shares, while Leisure occupies the smallest share.

Result:

The pie chart successfully represents the student's daily time allocation and makes comparison between activities easy.



P10 — BEAN PLOT: PLANT HEIGHTS

Aim:

To construct a bean plot and analyze the distribution and repeated values in plant-height measurements.

Data:

The plant heights are 12, 15, 15, 15, 18, 20, 22, 25, 25, and 30 cm.

Method:

The height values are plotted along a numerical scale. Individual observations are represented so that repeated measurements can be easily identified.

Analysis:

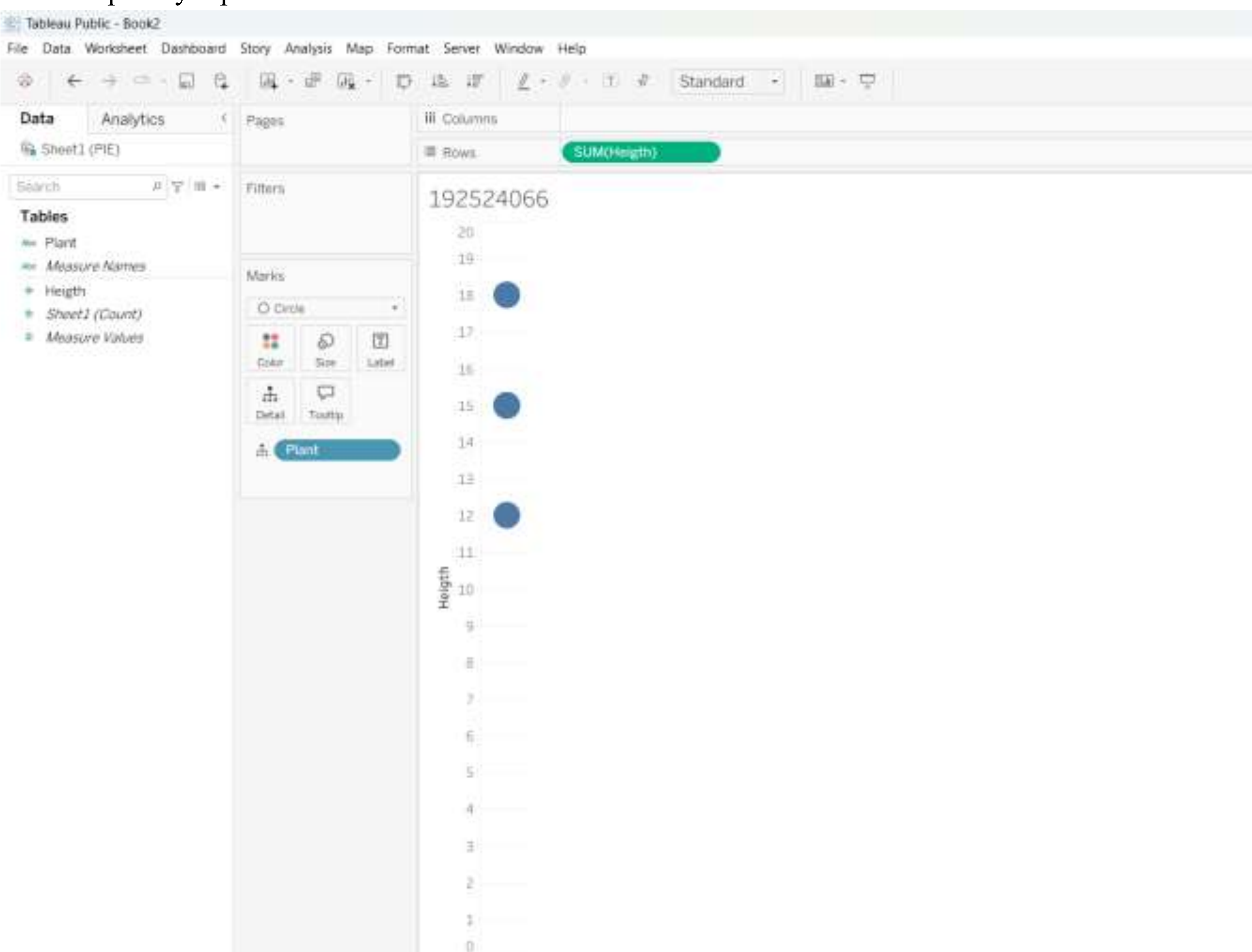
The minimum height is 12 cm and the maximum height is 30 cm. The value 15 cm occurs three times, while 25 cm occurs twice. The other values occur only once.

Interpretation:

The highest concentration of observations is around 15 cm. The repeated value at 25 cm also forms a smaller concentration. This shows that the plants did not all have unique heights.

Result:

The bean plot successfully displays the distribution of plant heights and clearly identifies 15 cm as the most frequently repeated measurement.



P11 — VIOLIN PLOT: EXAM SCORE DISTRIBUTION

Aim:

To construct a violin plot and analyze the distribution and concentration of students' examination scores.

Data:

The scores are 45, 50, 55, 58, 60, 62, 63, 65, 68, 70, 75, and 85 marks.

Method:

The examination scores are represented on a numerical scale. The width of the violin indicates the concentration of observations at different score levels.

Analysis:

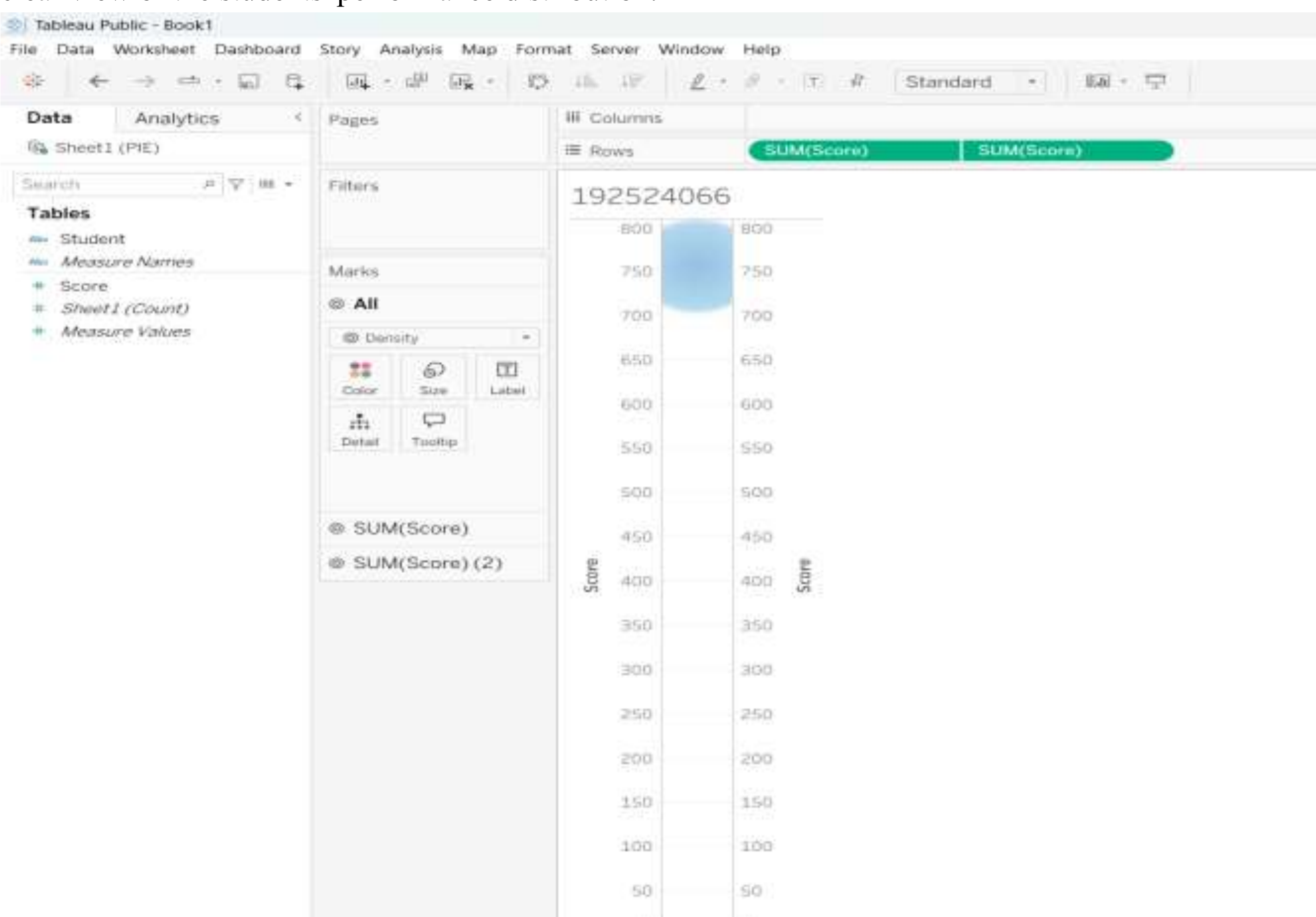
The scores range from 45 to 85 marks. Most of the observations are located between 55 and 75 marks. Several scores are closely grouped around 60–70 marks, while 45 and 85 are located toward the extremes.

Interpretation:

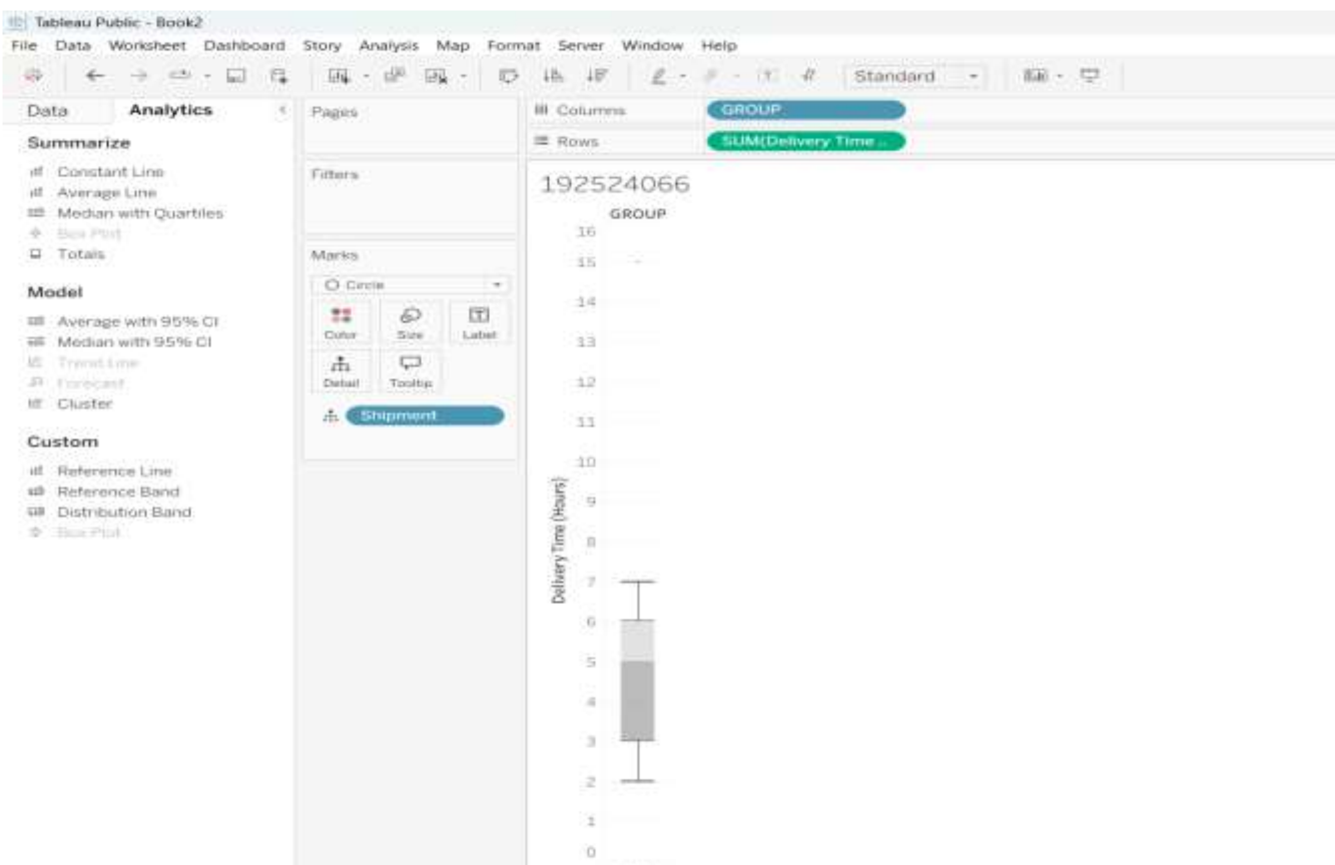
The central portion of the violin is expected to be wider because more observations are concentrated in that region. The thinner ends represent fewer observations.

Result:

The violin plot effectively represents the spread and concentration of examination scores and provides a clear view of the students' performance distribution.



P17 — BOX PLOT: DELIVERY TIME
Aim:
To construct a box plot for shipment delivery times and identify an outlier using the 1.5 × IQR rule.
Data:
The delivery times are 2, 3, 3, 4, 5, 6, 6, 7, and 15 hours.
Method:
The data is arranged in ascending order and the quartiles are calculated. Q1 = 3, Median = 5, and Q3 = 6.5.
Calculation:
IQR = Q3 − Q1
= 6.5 − 3
= 3.5
Upper limit = Q3 + 1.5 × IQR
= 6.5 + 1.5 × 3.5
= 11.75
Since 15 is greater than 11.75, the value 15 hours is an outlier.
Interpretation:
Most shipments were delivered within 2 to 7 hours. The 15-hour delivery is considerably higher than the normal range and is therefore represented separately as an outlier in the box plot.
Result:
The box plot successfully summarizes the delivery-time distribution and identifies 15 hours as the outlier.



P23 — STACKED BAR CHART: COMPANY HEADCOUNT

Aim:

To construct a stacked bar chart to compare company headcount and departmental composition for 2022 and 2023.

Data:

In 2022, Engineering has 40 employees, Sales has 25, and Support has 15. In 2023, Engineering has 55, Sales has 30, and Support has 20 employees.

Method:

Year is represented on the horizontal axis, while headcount is represented on the vertical axis. Engineering, Sales, and Support are shown as separate sections of each stacked bar.

Analysis:

The total headcount in 2022 is 40 + 25 + 15 = 80 employees. The total headcount in 2023 is 55 + 30 + 20 = 105 employees. Therefore, the workforce increased by 25 employees.

Interpretation:

Engineering has the highest contribution in both years. Sales and Support also show an increase in 2023. The complete height of each stacked bar represents the total workforce.

Result:

The stacked bar chart clearly shows the increase in company headcount from 80 employees in 2022 to 105 employees in 2023, along with the contribution of each department.